



School of Information Technology, Engineering, Mathematics and Physics

**CS341 – Software Quality Assurance and Testing
Semester 2, 2023
Mode: Face to Face**

**Mid-Semester Test
Date: Thursday 21 September 2023, Time: 11.00 – 11.50 am (FJT)**

*Time Allowed: 50mins
30 marks (Weight: 10%)*

Student ID: _____
Student Name: _____

LEARNING OUTCOMES:

- Explain the terminology and principles associated with software quality assurance and testing.

INSTRUCTIONS

- 1) This test has two sections:
 - a) Section A: Short Answer Questions (15 marks)
 - b) Section B: Long Answer Questions (15 marks)
- 2) Answer all questions in this booklet. All questions are compulsory.
- 3) The total number of pages including this cover sheet is 5.

~ALL THE BEST~

Section A: Short Answer Questions

(15 Marks)

1. Differentiate between Black box and white box testing (2 marks).

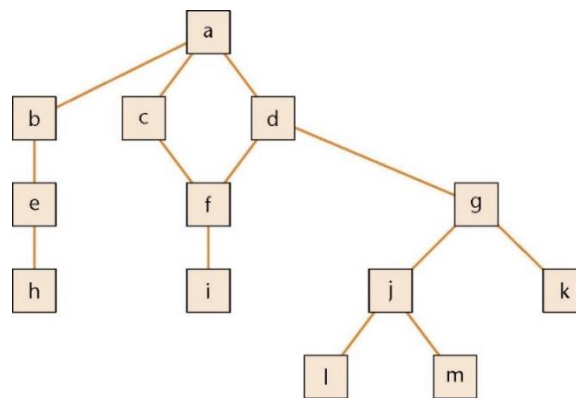
Black box testing

- Testing that ignores the internal mechanism of the system or component and focuses solely on the outputs in response to selected inputs and execution conditions.
- Testing conducted to evaluate the compliance of a system or component with specified functional requirements.

White box testing

Testing that takes into account the internal mechanism of a system or component.

2. Consider the following system with 13 modules.



- a) List the way you can integrate this system using the top-down integration approach, i.e write the sequence of modules you will integrate using the top-down integration method. (2marks)

a, b, c, d, e, f, g, h, i, j, k, l, m

- b) Give one advantage of this method. (1 marks)

In the Top-Down Integration testing approach, the main module is designed at first then the submodules/subroutines are called from it. *Ensure an example is used to explain.

- Fault isolation
- Stubs are not wasted
- Major design faults show up early

3. Review of work will most definitely be a usual occurrence for a team of 10 developers. Looking closely at Code Reviews explain what are Code Walkthroughs and Desk Checks. (2 marks)

Code Walkthrough is a form of peer review in which a programmer leads the review process and the other team members ask questions and spot possible errors against development standards and other issues. The meeting is usually led by the author of the document under review and attended by other members of the team.

A desk check is an informal non-computerized or manual process for verifying the programming and logic of an algorithm before the program is launched. A desk check helps programmers to find bugs and errors which would prevent the application from functioning properly.

4. A program has three logical operations for type of sugar level are shown below:

Normal	Prediabetic	Diabetic
Less than 5	Between 5.1 and 8	More than 8.1

*Assume the argument “type” is of type sugar level and the method returns “X” for invalid level.
Maximum Sugar level can be 20.

After implementing the method you are required to create unit test cases before your pull request can be processed. Using partition testing and equivalence partitioning method, which inputs will you pass in the test cases to cover all the valid and invalid partitions? (3 marks)

Valid: $\text{Sugar} \leq 5$, $5.1 < \text{Sugar} \leq 8$, $\text{Sugar} > 8.1$,
Invalid: $0 < \text{Sugar} > 20$

5. When an application is ready for launch there are of course a series of tests which are done. Generally, there are two streams of testing carried out, namely Alpha and Beta Testing. A group of USP students have create a basic lecture scheduling application. The application allows users to check of lecture locations, schedules as well as booking slots for consultations with their respective lecturers.

Use examples and describe how the respective testing streams can be carried out and who would be the testers in each stream. Also explain the differences between the two streams here. (2 marks)

Testing is done by customers or representatives of your market. This testing is done when the software is stable. The product is used in a realistic way in its operational environment. Both testing streams helps to: Give comments back on the product

- Faults found.
- How the product meets their expectations.
- Improvement / enhancement suggestions?

6. Measurement is the process by which numbers or symbols are assigned to attributes of entities (objects) in the real world in such a way as to ascribe them according to defined rules. This conforms to the world of software development and testing as well. In our learning journey this semester, much discussion was done on the levels of measurements. Determine and define in the table below, which scale should be used to measure the following scenarios: (3 marks)

You are conducting a survey to gather information about people's favorite types of music. Participants are asked to choose their preferred music genre from a list.	Nominal
Measuring attitude towards Internet services, if the evaluation scores are numerically meaningful so the difference between a rate of 3 and a rate of 6 is exactly the same as the difference between a rate of 7 and a rate of 10.	Interval
Measuring attitudes towards an Internet service (say, on an n-point rating, $n = 0$ to 10).	Ordinal

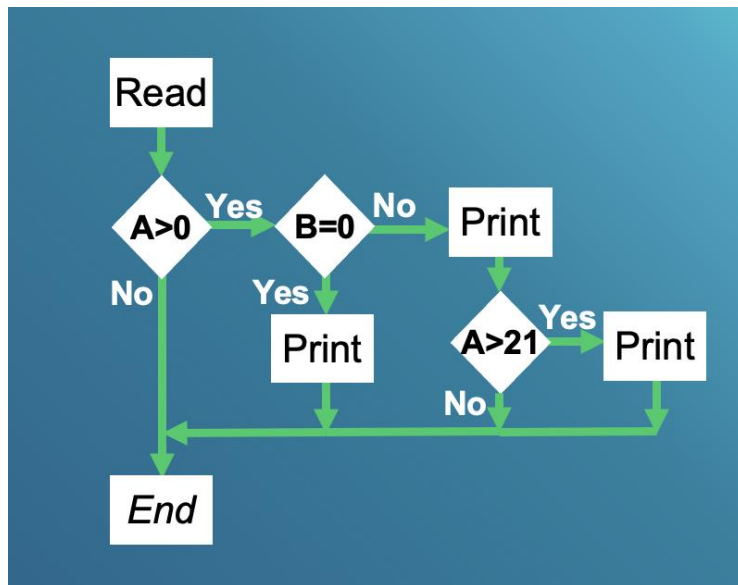
Section B: Long Answer Questions**(15 Marks)**

7. There are many 2 distinct branches when it comes to software testing, mainly known and white box and black box testing. An application development team is given a set of instructions or basically pseudocode to use to develop an application. Post development the development team have been charged with the task of answering questions such as how confident they are, that the tests they have done is sufficient enough to deem that the deliverable is of good quality.

Pseudocode:

```
Read A
Read B
IF A > 0 THEN
    IF B = 5 THEN
        Print "No values"
    ELSE
        Print B
        IF A > 10 THEN
            Print A
        ENDIF
    ENDIF
ENDIF
```

- a) For the pseudocode given above create the flow graph. (Mark 5)



b) Determine how many tests would be needed to perform: (2 marks)

i. Maximum Statement Coverage.

2

ii. Maximum Branch/Decision Coverage.

4

c) Calculate the cyclomatic complexity of the application and explain what significance does this measure have (2 marks).

Cyclomatic complexity = $E - N + 2 \cdot P$ where, E = number of edges in the flow graph. N = number of nodes in the flow graph. P = number of nodes that have exit points

4 – Answer. $(10 - 8 + 2(1))$

d) Define what is a Data Flow Anomaly and give an example from the above given pseudocode. (2 marks)

Data Flow Anomaly - n is re-defined without being used.

Data Flow anomalies are inconsistencies in the data defined or stored. Such inconsistencies may arise when have a particular record stored in multiple locations and not all of the copies are updated or are even being used.

8. Apply Decision Table Test Design technique for the following scenario. You are tasked with testing a login system for a website. The system allows users to log in with a username and password with account status being active. If any of the input is wrong, an error message will be displayed on the screen (4 marks).

Rule	Valid Username	Valid Password	Account Status	Allow Login	Display Error Message
R1	Yes	Yes	Active	Yes	No
R2	Yes	Yes	Inactive	No	Yes
R3	Yes	No	Active	No	Yes
R4	No	Yes	Active	No	Yes
R5	No	No	Active	No	Yes
R6	Yes	Yes	N/A	No	Yes
R7	No	No	Inactive	No	Yes
R8	No	N/A	N/A	No	Yes

THE END